EXP-3

STAIR CASE WIRING

AIM

To control the status of the given lamp by using two–way switches.

APPARATUS REQUIRED:

S. No.	Name of the Apparatus	Range / Type	Quantity
1	Incandescent Lamp	60 W	1 No.
2	Lamp Holder	Pendant Type	1 No.
3	SPDT Switch	230 V, 5 A	2 Nos.
4	Wires	1/18"	As per requirement
5	PVC Pipe	1/4"	As per requirement
6	Wooden Board	—	1 No.
7	Round Block	—	1 No.

Theory:

• A two way switch is installed near the first step of the stairs. The other two way switch is installed at the upper part where the stair ends.

• The light point is provided between first and last stair at an adequate location and height if the light is switched on by the lower switch. It can be switched off by the switch at the top or vice versa.

• The circuit can be used at the places like bed room where the person may not have to travel for switching off the light to the place from where the light is switched on.

• Two numbers of two-way switches are used for the purpose. The supply is given to the switch at the short circuited terminals.

• The connection to the light point is taken from the similar short circuited terminal of the second switch. Order two independent terminals of each circuit are connected through cables

PROCEDURE

• Place the accessories on the wiring board as per the circuit diagram.

• Place the P.V.C pipe and insert two wires into the P.V.C pipe.

• Take one wire connect one end to the phase side and other end to the middle point of SPDT switch 1

• Upper point of SPDT switch 1 is connected to the lower point of SPDT switch2.

• Lower point of SPDT 1 is connected to the upper point SPDT switch2.

• Another wire taken through a P.V.C pipe and middle point of SPDT switch 2 is connected to one end of the lamp holder.

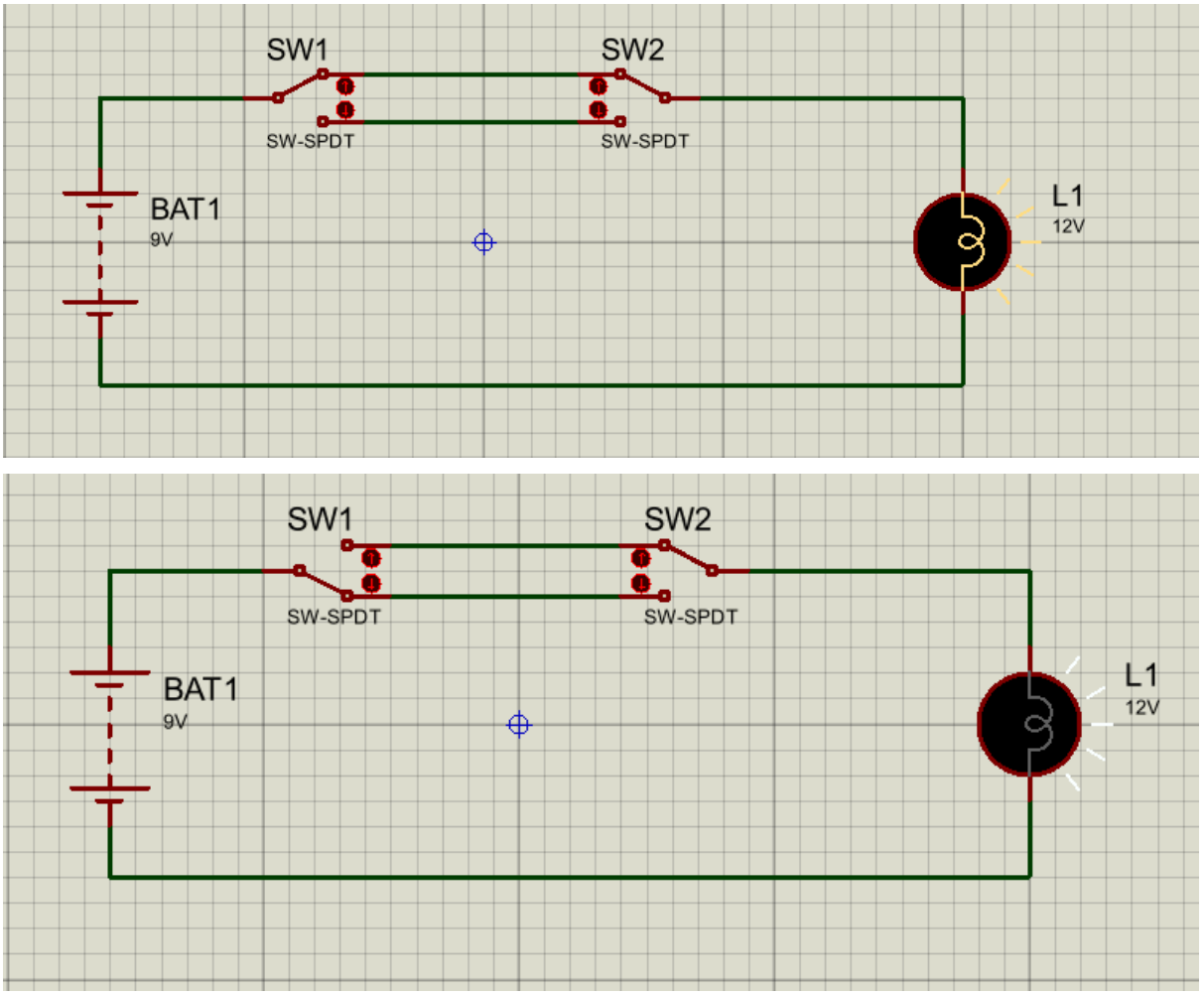
• Another end of lamp holder is connected to neutral line.

• Screw the accessories on the board and switch on the supply.

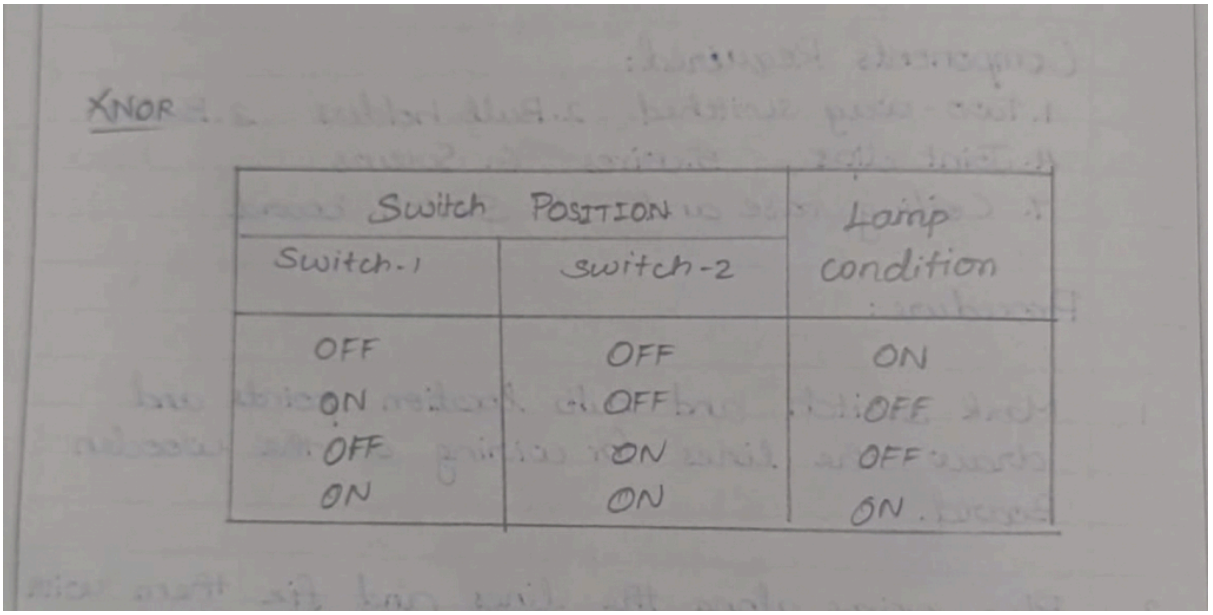
• Circuit is tested for all possible combination of switch positions.

Direct connection (XNOR):

CIRCUIT DIAGRAM:



Tabulation:1

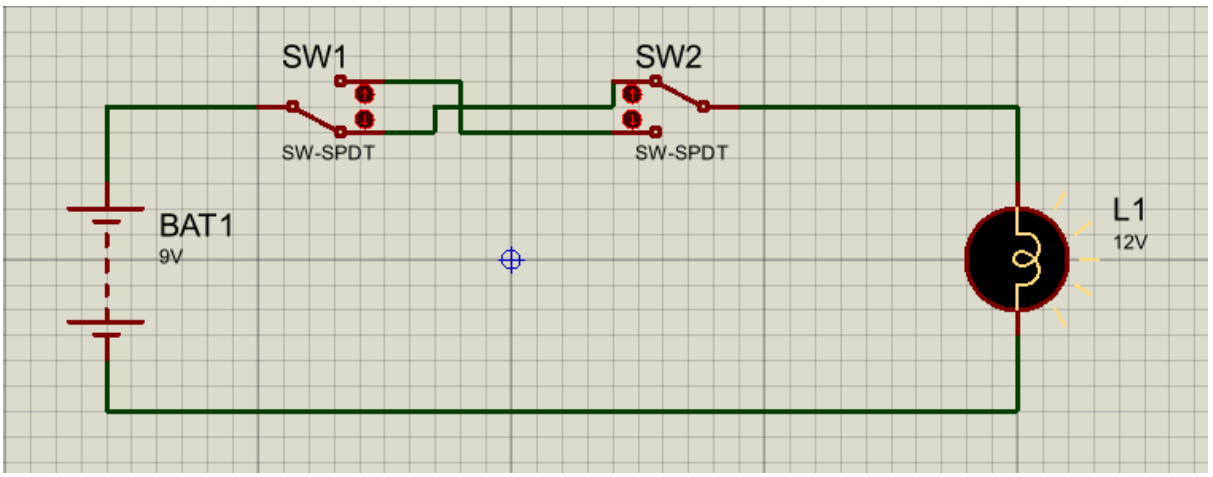


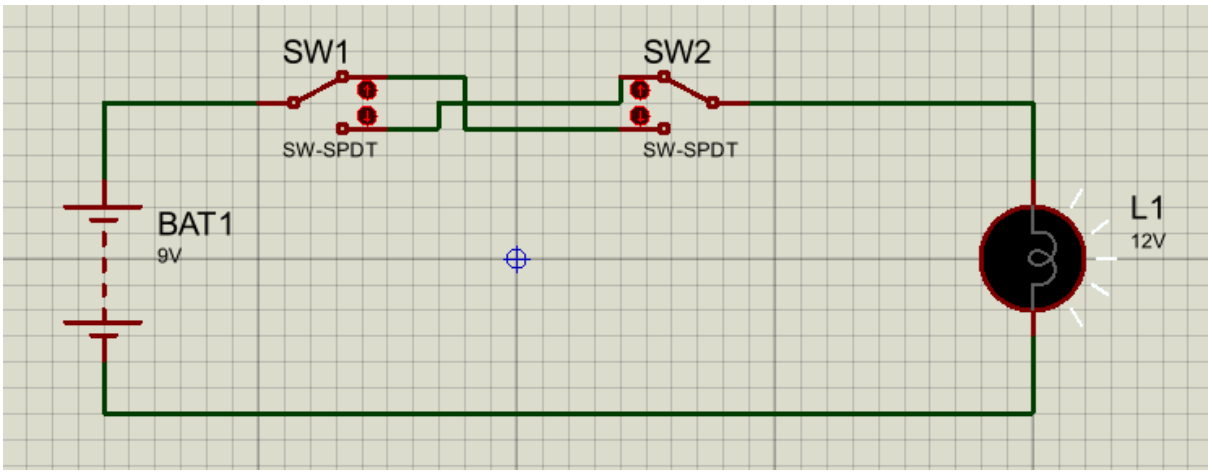
XNOR

Switch POSITION		Lamp condition
Switch-1	switch-2	
OFF	OFF	ON
ON	OFF	OFF
OFF	ON	OFF
ON	ON	ON

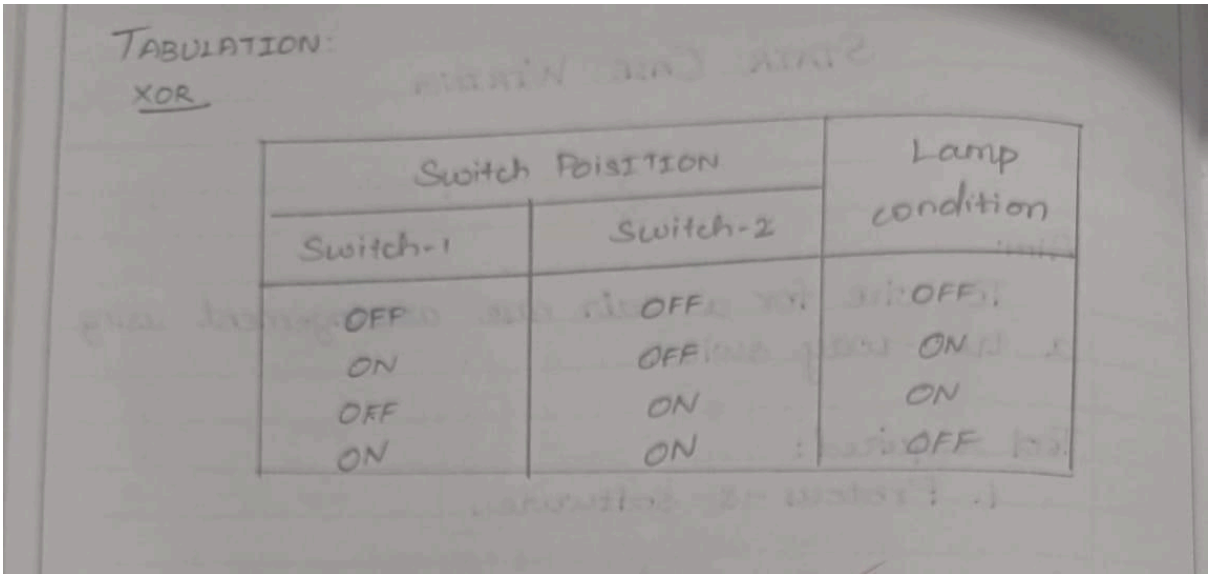
Cross connection (XOR) :

CIRCUIT DIAGRAM:





Tabulation:2



TABULATION:

XOR

Switch Position		Lamp condition
Switch-1	Switch-2	
OFF	OFF	OFF
ON	OFF	ON
OFF	ON	ON
ON	ON	OFF

RESULT:

Thus the staircase wiring is connected and tested.